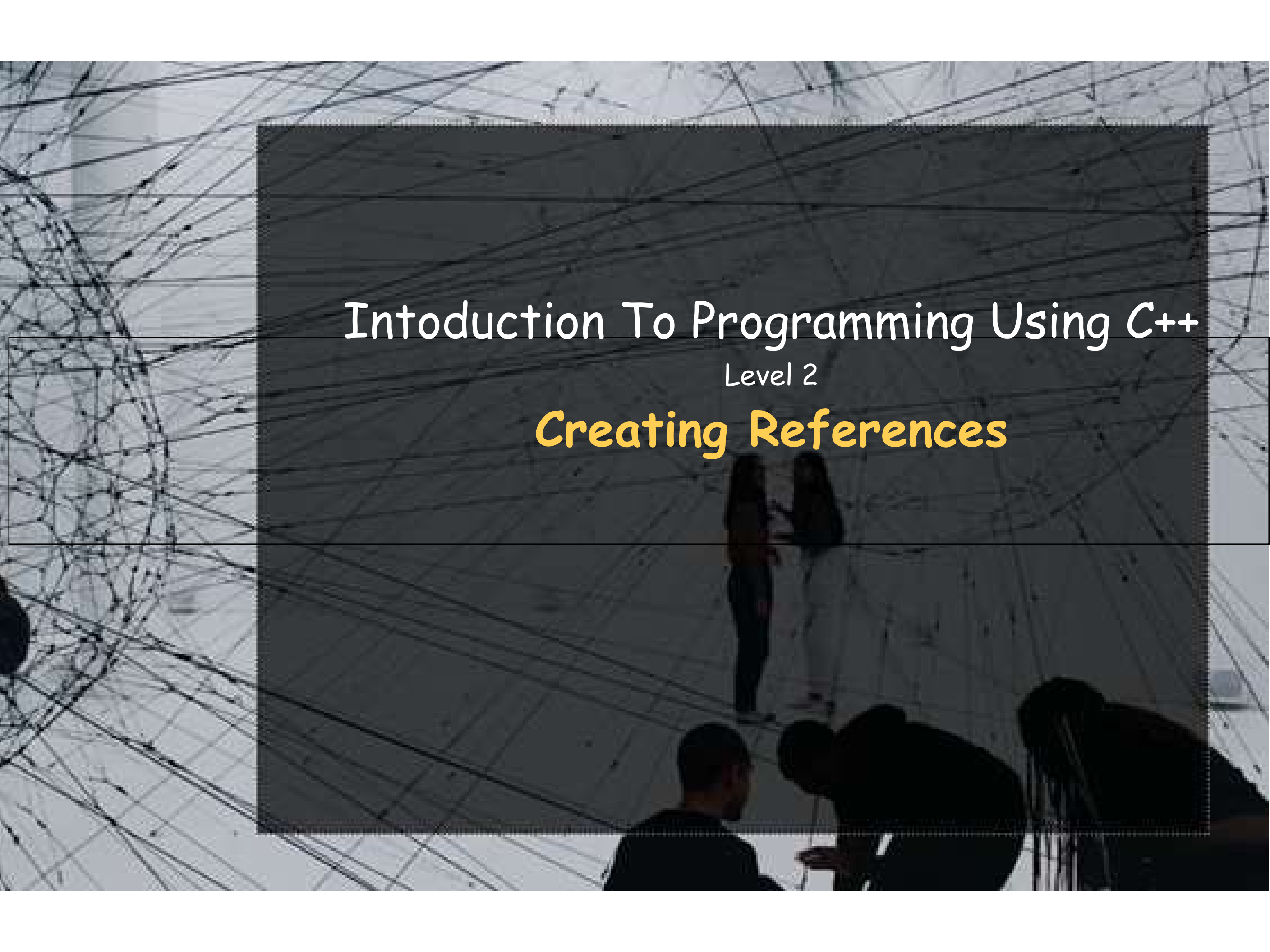




Introduction To Programming Using C++ Level 2

Creating References



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Creating References

Definition

Reference

-  A Reference is another name (Alias) for an existing variable.

-  A Reference refers to the same variable in memory and does not create a separate variable.

Explanation

Creating a Reference

-  A reference is created using the & symbol.
-  A reference must be initialized when it is declared.
-  Once a reference is bound to a variable, it cannot be changed to refer to another variable.

Procedures

1. Create a variable in main().
2. Assign a value to the variable.
3. Create a reference using the & symbol:
 `int& x = a;`
4. Use the reference to access the original variable.
5. Modify the reference.
6. Check the value of the original variable.
7. Modify the original variable.
8. Check the value through the reference.

💡 Example

◆ Creating a Reference

```
int main()
{
    int a = 10;
    int& x = a;
    cout << &a << endl;
    cout << &x << endl;
    cout << a << endl;
    cout << x << endl;
    x = 20;
    cout << a << endl;
    cout << x << endl;
    a = 30;
    cout << a << endl;
    cout << x << endl;
    return 0;
}
```

💡 Example

➡ Result

Initially:

$a = 10$

$x = 10$

After $x = 20$:

$a = 20$

$x = 20$

After $a = 30$:

$a = 30$

$x = 30$

Note

◆ A reference must be initialized when it is declared:

```
int& x = a;
```

◆ The following is incorrect because the reference is not initialized:

```
int& x;
```

◆ A reference cannot be null because it must refer to an existing variable or object.

◆ A reference cannot be changed later to refer to another variable.

◆ You can use &Name with cout to print the memory address of a variable:

```
cout << &Name << endl;
```

⊘ Warning

◆ Do not confuse the & used for references with the & used to get a memory address.

◆ In:

```
int& x = a;
```

▨ & is used to declare x as a reference.

◆ In:

```
cout << &a << endl;
```

▨ & is the address-of operator and returns the memory address of a.



Comparison

Reference

- ✓ Acts as another name (Alias) for an existing variable.
- ✓ Must be initialized when declared.
- ✓ Cannot be null or changed to refer to another variable.

Pointer

- ✓ Stores the memory address of a variable.
- ✓ Can be changed later to point to another variable.
- ✓ Can be null.

✦✦ Summary

- ✓ A Reference is another name (Alias) for an existing variable.
- ✓ A Reference refers to the same variable in memory and does not create a separate variable.
- ✓ Changing the reference changes the original variable, and changing the original variable is reflected through the reference.

Thank you 😊